



TEXAS A&M UNIVERSITY
ROBOTICS TEAM & LEADERSHIP EXPERIENCE

Materials, Manufacturing, GD&T

TURTLE MI Week 4

Topics Covered



- Basic Materials
- Manufacturing
- 3D-printing
- GD&T



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Materials

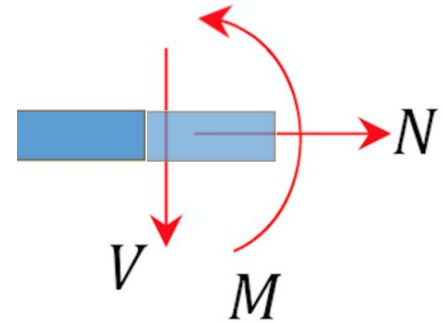
Internal Forces



There are 3 fundamental internal forces that a material has.

1. Normal force: perpendicular to cross section
2. Shear force: parallel to cross section
3. Bending moment: internal couple from loads that cause member to curve

(Shear and bending moment will be elaborated on in week 5)

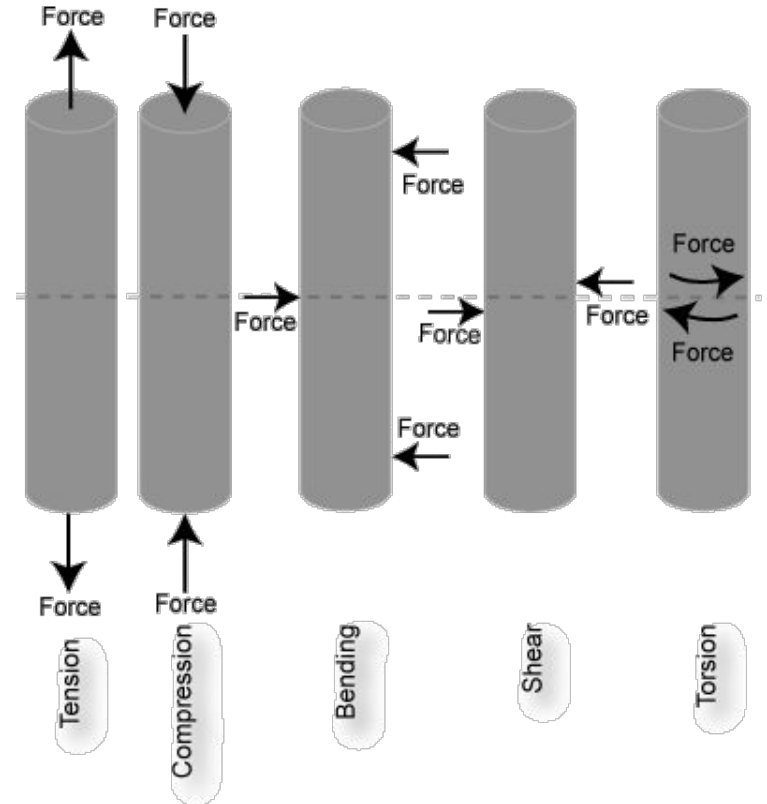


Loading Conditions



There are 5 loading conditions which are a combination of the 3 internal forces

1. Tension
2. Compression
3. Bending
4. Shear
5. Torsion



Loading Conditions



Definition: application of a force to an object

Tension: two sections of material on either side of a plane tend to be pulled apart/elongated

Compression: pressing the material together

Bending: applying a load in a manner that causes a material to curve

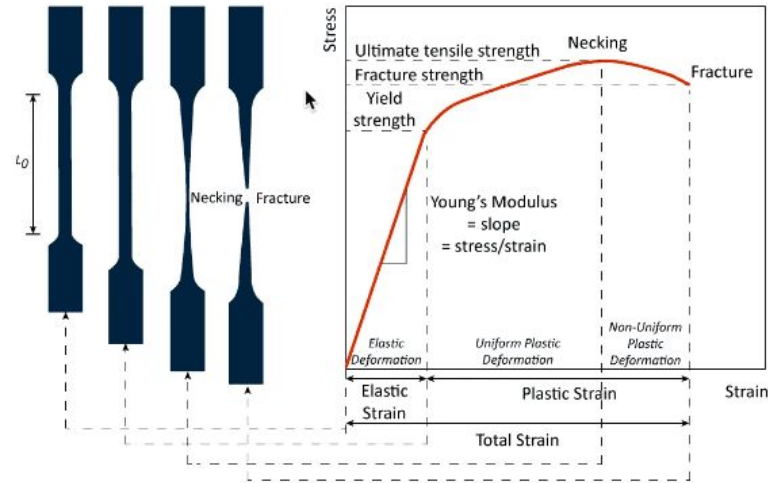
Shear: applying a load parallel to a plane

Torsion: application of a force causing twisting in a material

Stress-Strain Curves



- The Stress-Strain curve for a material gives the relationship between a material's applied pressure (stress) and its deformation (strain)
- This can provide valuable insight on many material qualities like Young's Modulus, Ductility, Tensile Strength, etc.



Stress Formula

$$\sigma = \frac{F}{A}$$

Strain Formula

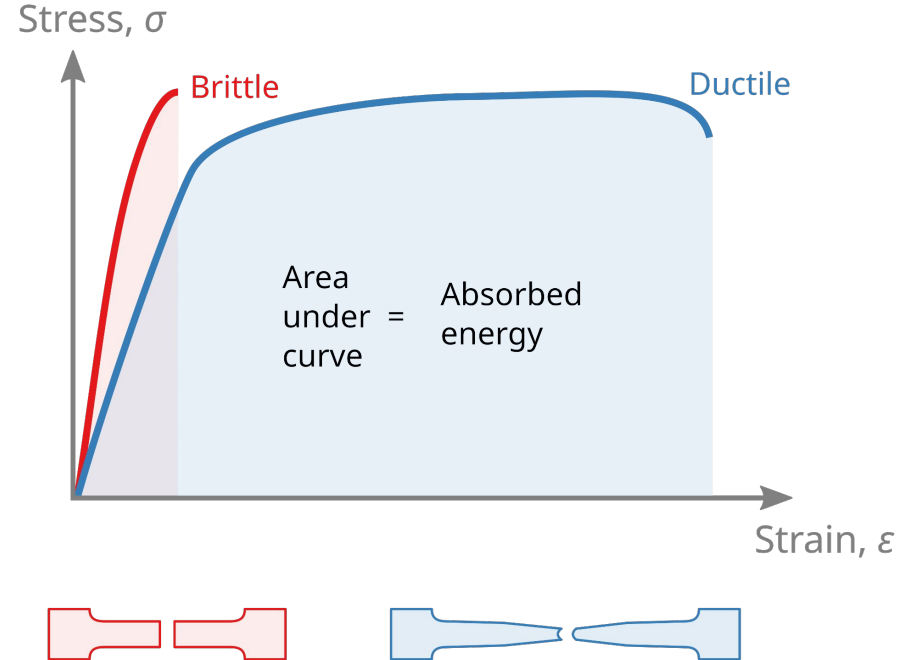
$$\varepsilon = \frac{\Delta L}{L}$$

- Stress: the internal force / unit area
- Strain: the measurable effect in size and shape

Ductile vs. Brittle



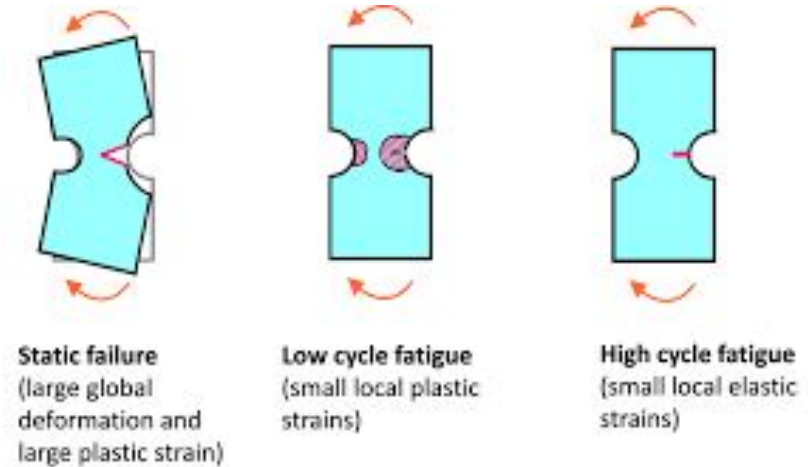
- Ductile materials can absorb lots of mechanical energy through impacts through plastic deformation
- Brittle materials usually will not yield to absorb energy and will shatter
- Metals and Plastics are usually more ductile while glasses and ceramics are brittle



Fatigue



- repeated loading/unloading may cause micro-cracks in the material which grow over time.
- This can be important to consider when designing for projects that involve cyclic loading or stresses over long period of time like bridges, pressure vessels, aircraft, etc.
- Creep Failure: permanent deformation of material under constant load and temperature
- Creep deformation: characterized by the elongation of a crack over a period of time
- Fatigue cracks propagate quickly in brittle materials causing them to shatter violently while ductile materials will usually yield instead



Fatigue Demo →

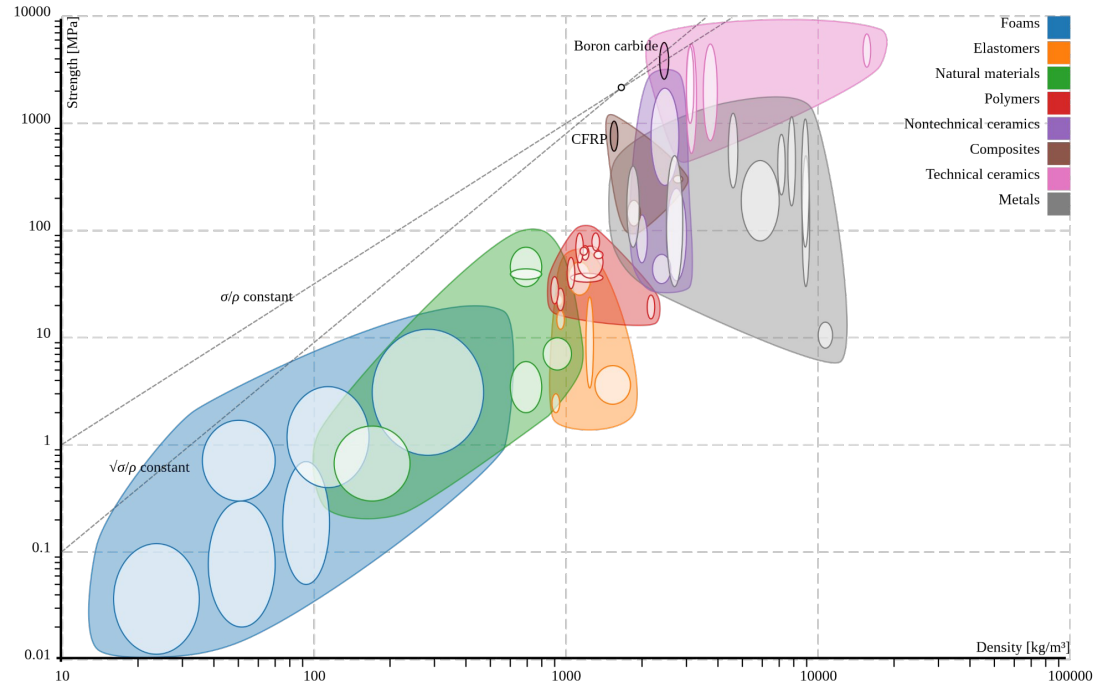
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Strength-to-Weight Ratio



- Also called Specific Strength, is a measure of strength in relation to weight, indicates the efficiency of used materials
- Is very relevant to **Antweight** combat robotics
- Examples of materials with high strength-to-weight ratio:
carbon fiber, titanium, aluminum, magnesium alloys, steel
- Can affect design decision making (ex. could choose steel compared to Al, Ti, Mg at a weight penalty)



Thermal Properties



- Materials can have many thermal properties that will affect how a material behaves when exposed to changes in temperature.
- Heat Capacity: Quantifies the heat required to raise the temperature of a material
(ex: water has a heat capacity of $c = 4.184 \text{ J/(g}^\circ\text{C)}$ -> it takes 4.184 Joules to heat up 1 gram of water by 1°C)
- Thermal Expansion: Describes the change in dimensions of a material in response to temperature variations, can cause Thermal Stress
- Thermal Conductivity: Measures the ability of a material to transfer heat through conductivity

-
- Diagram illustrating the Charpy impact testing machine setup:
- Gauge measures in joules (J)**: The gauge is positioned at the pivot point of the pendulum.
 - Pendulum start height**: The initial height of the pendulum arm before impact.
 - Pendulum finish height**: The height of the pendulum arm after impact.
 - Anvil**: The base on which the Charpy specimen is placed.
 - Charpy specimen 55 mm × 10 mm × 10 mm**: The specimen being tested, featuring a V-shaped notch.
 - Notch details**: The notch has a 45° angle, a 2 mm depth, and a 0.25 mm radius at the bottom.

Material Types



Metals & Alloys:

- + High Electrical/Thermal Conductivity
- + High Strength
- + High Stiffness
- + High Ductility
- + High Toughness



Polymers/Plastics:

- - Low Electrical/Thermal Conductivity
- - Low Strength
- - Low Melting Point
- - Low Stiffness
- + Cheap and easy to form



Ceramics & Glasses:

- - Low Electrical/Thermal Conductivity
- + High Melting Point
- + Strong in Compression
- + High Stiffness
- + High Hardness
- - Brittle



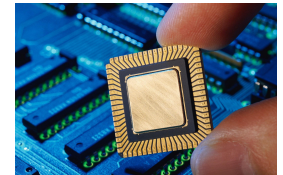
Composite Materials:

- Formed from 2 or more materials
- Can produce properties not found in a single material
- Usually have a preferred direction (strong in 1 plane or direction but weak in another)



(Semiconductors):

- Mixed Electrical Conductivity
- Useful in electronics





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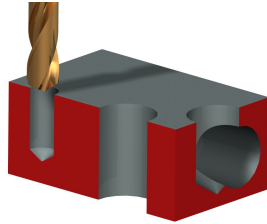
Manufacturing

Features

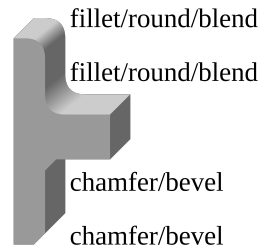


- Feature: distinct geometric element of a part with a specific purpose, created by a specific manufacturing operation.
- **Why are features important?:** every feature on a part needs to be producible by the manufacturing method you are using.

Hole



Fillet & Chamfer



Threads



Costs/Time



Adding unnecessary complexity to your project can dramatically increase the cost and time needed to complete it

Common things to look out for that can increase
Cost and/or Time:

- Exotic Materials
- Tight Tolerances
- Material Certifications
- Fragile Features
- Custom Tooling
- Complex Assembly
- Many Features
- Complex Geometry
- Multiple Processes required



While these features can be useful, it's important to consider their value in your final product

Manufacturing Types



There are many different ways to manufacture parts, each with its own pros and cons

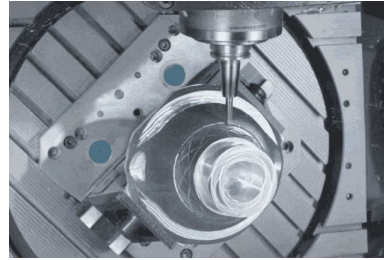
Casting



Process of pouring molten material into a mold to form a specified geometry
(Sand Casting, Die Casting, Investment Casting)

- +Rapid and cheap
- +Versatile
- +Complex Geometries
- Defects
- Lower strength
- Rough surfaces

Machining



Not limited to CNC machining but any process that uses a tool to remove material from a workpiece (Lathe, Milling, etc)

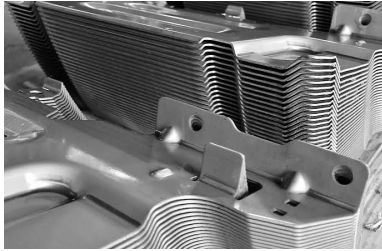
- +High precision
- +Fast for small batches
- +Versatile
- Material Wastage
- Needs a skilled operator
- Difficult to scale

Manufacturing Types



There are many different ways to manufacture parts,
each with its own pros and cons

Forming



Metal sheets can be formed by placing
force on specified areas
(Stamping, drawing, Hydroforming,
stretch forming Springback)

+90% of metals are
manufactured this way
+Cost effective, Fast

-Only net and near net
shapes
-High upfront cost
-Not Versatile

Extrusion



Non-ferrous metals can be
extruded to create fixed cross
sectional shapes

+Cost effective, Fast

-Limited to only uniform cross
sections

Additive Manufacturing (3D printing)

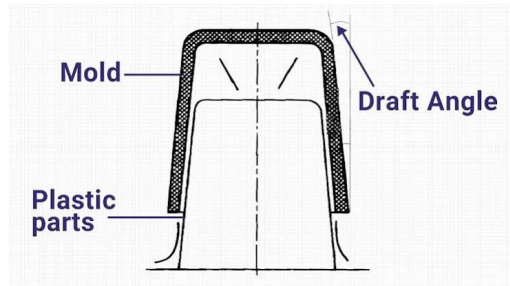


Design For Manufacturing & Assembly (DFMA)



It is important to consider processes, features, assembly, materials, tolerances to ensure everything goes together smoothly.

Some examples of things to think about when designing.



Some processes need special considerations like injection molding in order to be manufacturable



The designer for this part did not think about assembly and disassembly, so it is impossible to take apart



Just because a part is possible to make does not mean it is economical or viable to manufacture



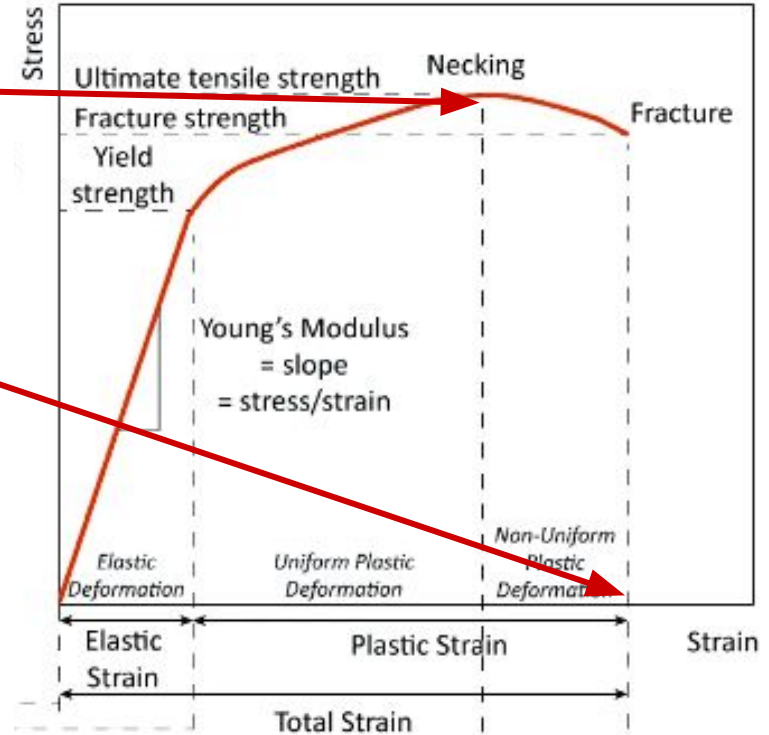
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3D Printing

Meaning of Properties



- Tensile Strength - maximum stress before necking and fracture
- Elongation at Break - how much the material stretches before fracture



Meaning of Properties



- Flexural Strength/Modulus - resistance to bending and stiffness under bending load
- Impact Strength - resistance to sudden shock loads

Flexural Modulus: $E_{\text{flex}} = \frac{L^3 F}{4wh^3 d}$

Flexural Strength: $\sigma = \frac{3FL}{2bd^2}$

Meaning of Properties

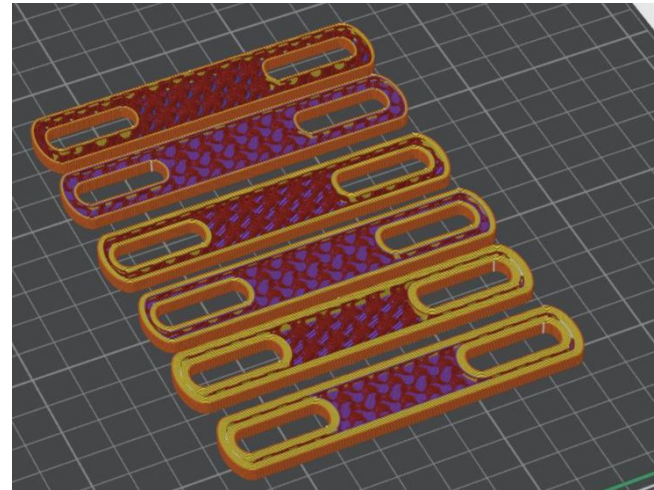
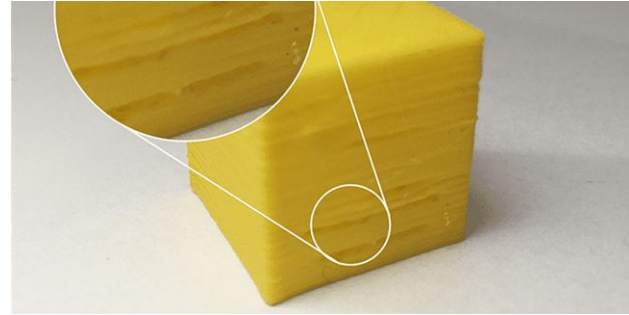


- Printing techniques govern properties much more than plastic material properties actually do
- Depending on the printing method, the part can keep between 50-90% of raw material properties. Properly optimized printing is extremely important in longevity and overall functionality.

Manipulating Mechanical Properties



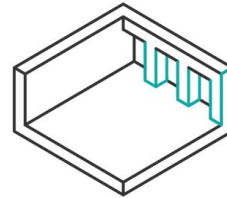
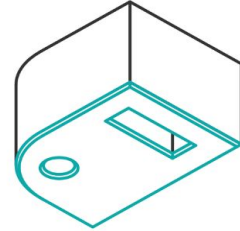
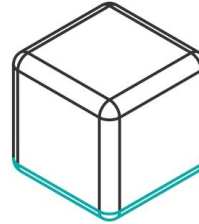
- Layer adhesion (z-axis strength)
 - Higher temps
 - Slower speed
- In-plane strength and stiffness
 - Wall thickness (better)
 - Infill % (worse)
- Toughness
 - Slower speed
 - Wall thickness
- Print orientation matters!



Best Practices



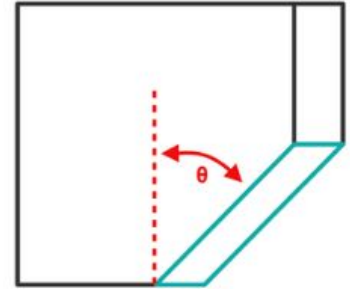
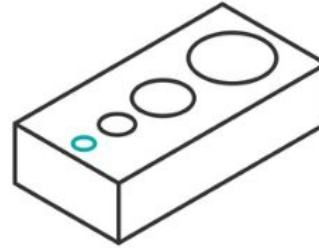
- **Fillets on sharp corners**
 - Minimize stress concentration
 - Improve bed adhesion if in contact with build plate
- **Base chamfers / brim**
 - Improve base edge accuracy
 - Choose one or the other, using both isn't necessary
- **Bridging**
 - Unsupported bridges shouldn't be longer than ~10mm or ~0.4in
 - Incorporate vertical structures into the part or use supports in the slicer
- **Feature size**
 - Recommended minimum is 4 times the nozzle diameter
 - Bare minimum is nozzle diameter, but not recommended



Best Practices



- Layer lines affect strength, so orient load-bearing parts so layers run perpendicular to the stress direction
- Hole size
 - Hole diameter should be at least 2 mm to ensure proper printing
- Overhangs
 - Unsupported overhangs should be printed at angles no greater than 50 degrees from the vertical
- Supports
 - Use supports when necessary for better print quality, strength, and overall proper practice

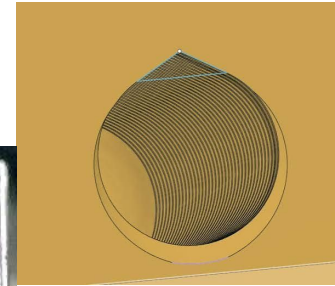


<https://www.hydraresearch3d.com/design-rules>

Holes



- Hole deformation
 - Avoid printing perpendicular to plate
 - If accuracy is critical, print holes parallel to plate
 - Add tear drop/ point to perpendicular holes
- Add tolerances to holes
 - Add 0.2 - 0.4 mm to diameter for bolts
 - 3D prints shrink as they cool



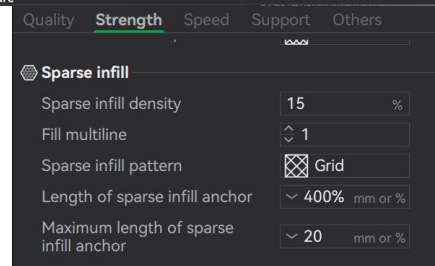
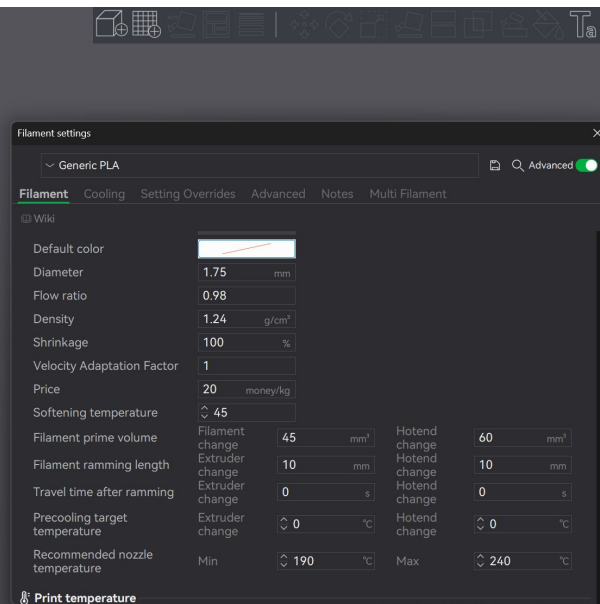
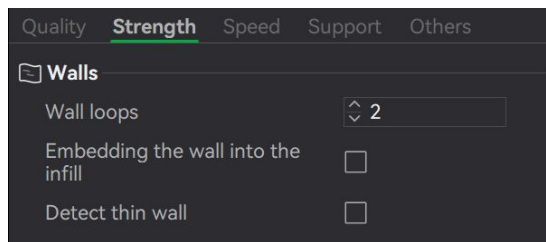
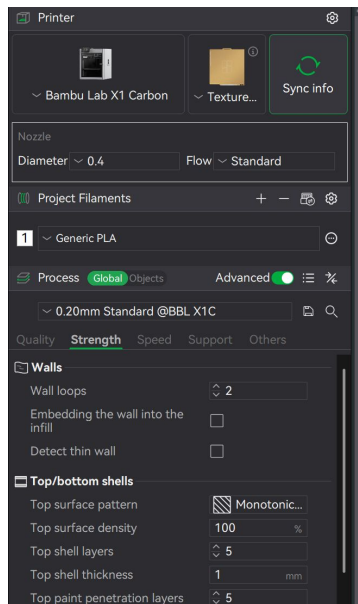
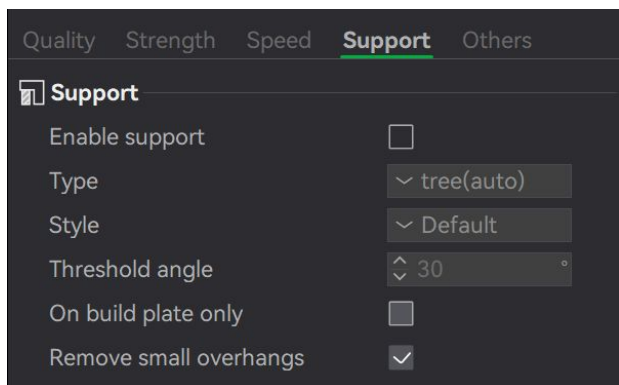
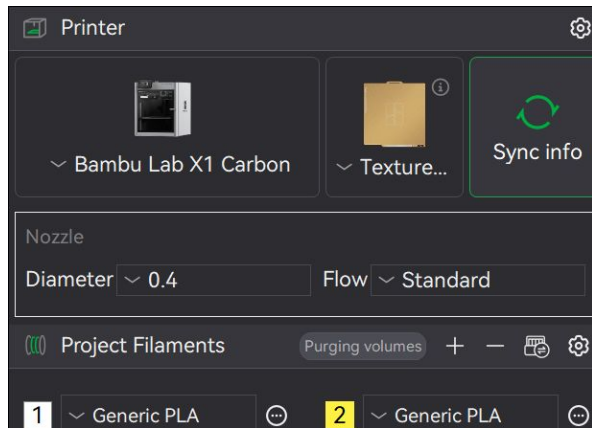
Slicer customization



Things you can change in the slicer

- Print speed
- Print temp
- Wall thickness / wall loop count
- Infill %
- Infill pattern
- Support types
- Nozzle size

Slicer customization



Battle Bot Examples

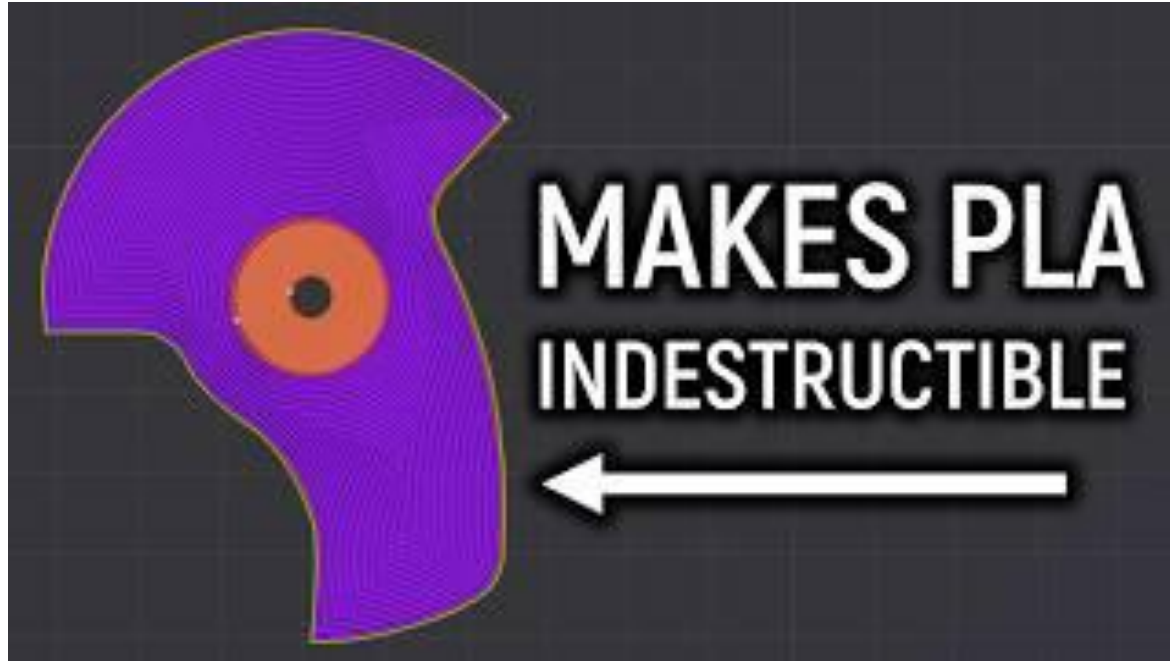


Violent Crumble

Dense print settings (SVRN armor):

- 4-6 perimeters
- 4-5 top/bottom layers
- $\geq 20\%$ infill for PLA-ST part

Battle Bot Examples





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GD&T

Symbols



- GD&T symbols are a way of defining allowable variation in form, orientation, location, and size of part features
- 5 categories:
 - Form (yellow)
 - Orientation (blue)
 - Location (tan)
 - Profile (orange)
 - Runout (green)

GD&T Symbols Chart

Straightness		True Position	
Flatness		Concentricity	
Circularity		Symmetry	
Cylindricity		Profile of a line	
Angularity		Profile of a Surface	
Perpendicularity		Total runout	
Parallelism		Circular runout	

Symbols

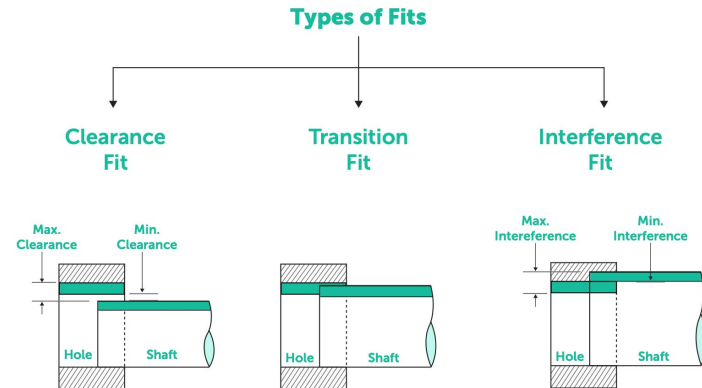
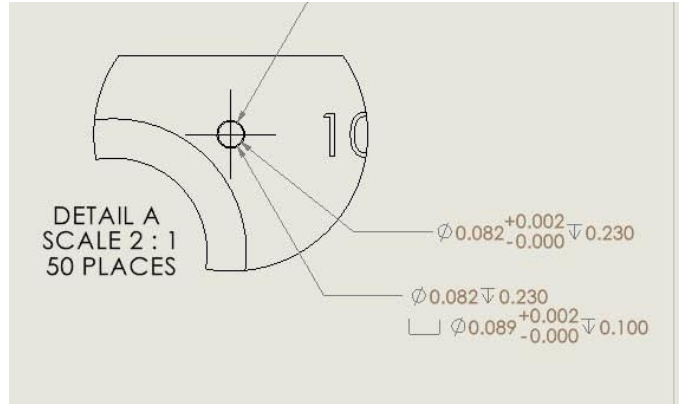


- Form
 - Controls the shape of a feature independent of other features
- Orientation
 - Controls how a feature is angled relative to a datum
- Location
 - Controls where a feature is located relative to a datum
- Profile
 - Can control form, orientation, and location depending on how it's applied
- Runout
 - Controls variation during rotation (common for shafts and rotating parts)

Standard Tolerances



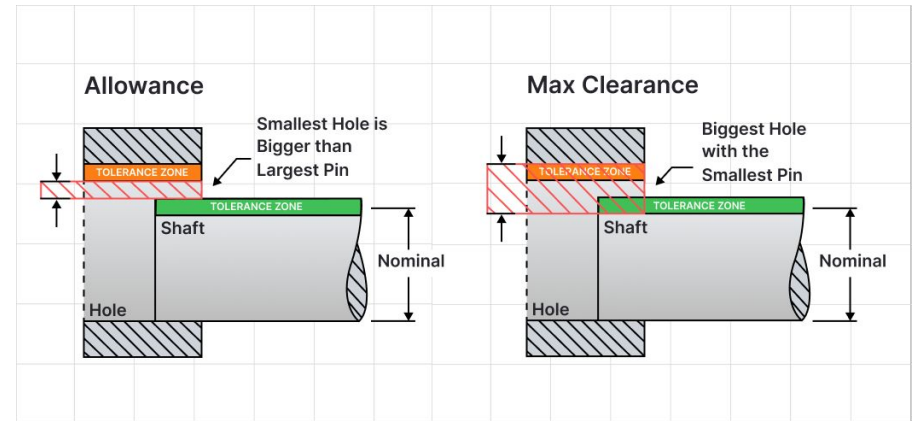
- Tolerance: The acceptable range of variation in a dimension
- It is important to call out your tolerances before manufacturing to ensure your parts fit together well

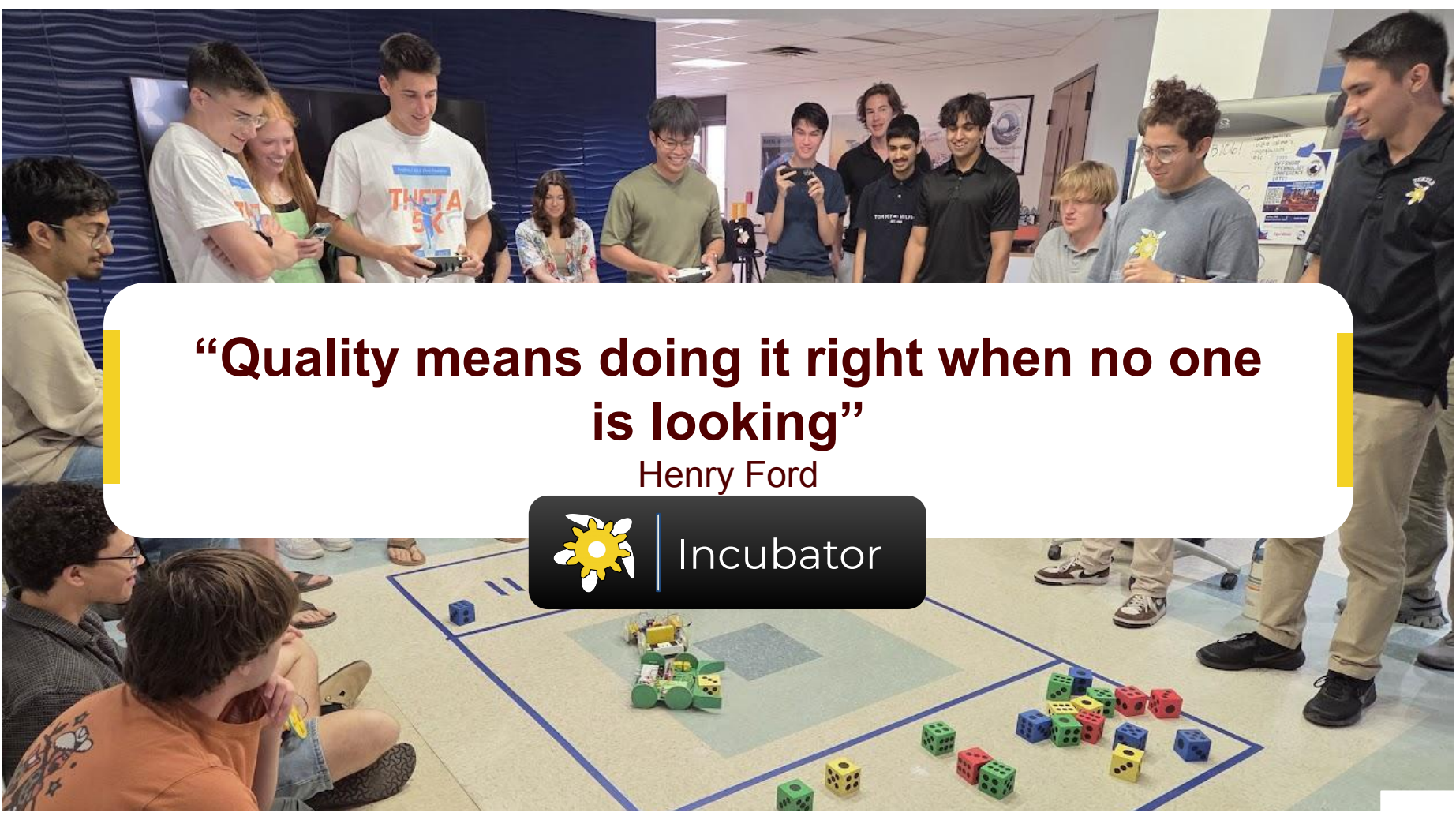


Tolerance Build-Up



- When multiple tolerances in a part or an assembly can add up to larger variations
- Really matters in fits, holes, or clearance regions





**“Quality means doing it right when no one
is looking”**

Henry Ford



Incubator